

Where expansion comes from

inner arrows, outer symmetries, Galton–Watson duality,
and a neural Gabber–Galil lemma

Working note IV for the continuation-algebra programme
self-reviewed draft, not independently verified

3 October 2026

Abstract

Note III proposed a *neural Gabber–Galil lemma*: expanding, measure-preserving transformation families on what a network distinguishes, with a certificate built from weight data. Developing it forced a classification of where expansion can come from on a Cantor hierarchy. We use three external inputs: topological full groups and their amenability (Matui’s survey; Juschenko–Monod), the measured orbit-equivalence category and its cohomological *shadow* (Medynets–Sauer–Thom, MST), and tiling deformations (Julien–Sadun, JS).

(1) Inner arrows never expand. For the tail/continuation relation (AF) and for minimal $\mathbb{Z}$ -actions, the topological full group is amenable. No finite set of inner symmetries, i.e. arrows of the continuation algebra assembled into homeomorphisms, gives a spectral gap. Bounded-degree uniform expanders on the cylinders are impossible from the convolution algebra’s own arrows. This is why notes I and III needed per-level generators.

(2) Outer symmetries do, and rank two is where they start. Interleaving digits identifies the fair binary Cantor set with the 2-adic plane $\mathbb{Z}_2^2$. Sanov’s free group $\langle (\begin{smallmatrix} 1 & 2 \\ 0 & 1 \end{smallmatrix}), (\begin{smallmatrix} 1 & 0 \\ 2 & 1 \end{smallmatrix}) \rangle \cong F_2$ then acts by measure-preserving homeomorphisms that preserve *every* cylinder partition and normalize the tail relation. Their MST shadows are the matrices themselves; their JS type is a pure linear deformation. Every finite level is a Gabber–Galil expander (second eigenvalue $\leq 5\sqrt{2}/8$). For a single odometer the shadow lies in $\{\pm 1\}$, which is why the Gabber–Galil mechanism needs two digit streams.

(3) The local lemma is a frequency-drift criterion, and its network analogue is Galton–Watson criticality. We prove a Schur–Kesten criterion that turns outward drift of the dual action on frequencies into a spectral gap; Gabber–Galil’s local lemma is an instance. For deep random networks the dual action on moment orders is a Galton–Watson branching. Subcritical (ordered) networks forget the input exponentially. Supercritical (chaotic) networks fool every fixed-degree test exponentially and act as pseudorandom generators. He-ReLU is exactly critical, with only polynomial (D^{-2}) behaviour; this is confirmed numerically for tanh in both phases and for ReLU.

(4) Neural Gabber–Galil lemma (walk form). Lump the interleaved Gabber–Galil walk onto *any* partition of inputs induced by the network (continuation classes, ε -chain classes of any layer, activation tiles) and Metropolize toward the gain-weighted class law. The result is a reversible walk on the network’s own distinctions, with certified spectral gap at least $(1 - 5\sqrt{2}/8) \cdot \min h / \max h$, where h is the class-averaged gain. On a random ReLU network the lumped gaps are 0.24–0.30 (source 0.21) and the gain-targeted gaps are 0.23–0.30, above the certified 0.15–0.16.

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1 The symmetry layer: inner, outer, shadow

Throughout, X is a Cantor set with a profinite structure (PB Def. 7), i.e. nested finite clopen partitions $\mathcal{P}_m$. μ is a non-atomic invariant probability of full support, R is an étale equivalence relation preserving μ , and E_m is the μ -conditional expectation onto $\mathcal{P}_m$ -measurable functions.

Definition 1.1. • *Inner symmetries:* the topological full group $[[R]]$, i.e. homeomorphisms whose graph is a finite union of compact-open bisections of R (Matui, Def. 4.1). These are the arrows of the convolution algebra $C^*(R)$ assembled into maps.

- *Outer symmetries:* homeomorphisms h with $(h \times h)(R) = R$ (equivalently automorphisms of $[[R]]$, by Medynets’ reconstruction; MST Thm. 2.1), taken μ -preserving.
- *Shadow:* for a free minimal $\mathbb{Z}^d$ -action generating R , MST’s functor $\Psi_1 : \text{Aut}_\mu([X, \mathbb{Z}^d]) \rightarrow \{A \in GL_d(\mathbb{R}) : \det A = \pm 1\}$ (MST Thm. 4.7). For tiling hulls, JS’s cohomological deformation invariant plays the same role.

For a finite symmetric family S of μ -preserving homeomorphisms, put $P_S = |S|^{-1} \sum_{s \in S} U_s$ with $U_s f = f \circ s$.

Proposition 1.2 (uniform level gaps = global gap). *If every $s \in S$ maps each $\mathcal{P}_m$ to itself, then P_S commutes with every E_m . The spectral bound of P_S on $L_0^2(X, \mu)$ is the supremum of its spectral bounds on the level spaces $L^2(\mathcal{P}_m) \ominus \mathbb{C}$, whose union is dense.*

Proof. P_S is self-adjoint and preserves each finite-dimensional $L^2(\mathcal{P}_m)$, hence commutes with its projection E_m . Rayleigh quotients over a dense union of invariant subspaces determine the top of the spectrum. $\square$

2 Inner arrows never expand

Theorem 2.1 (inner no-go). *Suppose $[[R]]$ is amenable. This holds when R is AF (in particular the tail relation and every continuation relation, where $[[R]]$ is locally finite), and when R is the orbit relation of a minimal homeomorphism (Juschenko–Monod; Matui Thm. 8.4). Then for every finite symmetric $S \subset [[R]]$, P_S has no spectral gap on $L_0^2(X, \mu)$. Consequently the level Schreier graphs of S are not uniform expanders.*

Proof. For AF relations, compactness shows each $g \in \llbracket R \rrbracket$ acts at some level n as a permutation of level- n prefixes inside continuation classes. So $\llbracket R \rrbracket = \bigcup_n \prod_{C \in P_n} \text{Sym}(C)$ is locally finite, hence amenable. Let $\Gamma = \langle S \rangle$, an amenable group. If Γ is not ergodic, a nonconstant invariant $f \in L_0^2$ is an eigenvector with eigenvalue 1. If it is ergodic, then since μ is non-atomic and Γ amenable, the action is not strongly ergodic (Connes–Weiss; Schmidt). So there are sets A_k with $\mu(A_k) \in [c, 1-c]$ and $\mu(sA_k \triangle A_k) \rightarrow 0$ for all $s \in S$. Put $f_k = \mathbf{1}_{A_k} - \mu(A_k)$. Then $\langle f_k, (I - P_S)f_k \rangle = \frac{1}{2|S|} \sum_s \|U_s f_k - f_k\|^2 = \frac{1}{2|S|} \sum_s \mu(s^{-1}A_k \triangle A_k) \rightarrow 0$ while $\|f_k\|^2 \geq c(1-c)$. The last claim follows from Proposition 1.2. $\square$

Remark 2.2 (what this explains). Note I’s small-bias sparsifier used $O(N)$ jumps per level and note III’s tail operators used eight maps *per level*. Theorem 2.1 shows that some such growth is forced as long as the generators are inner. The fair PB hierarchy is an odometer (note I, Thm. 9.3), so its inner symmetries are amenable twice over.

Remark 2.3 (the semigroupoid is non-amenable but has no invariant measure). The shift semigroupoid of note I, §7 has groupoid $G_{[2]}$ with $C^*(G_{[2]}) = \mathcal{O}_2$. Its full group is the Higman–Thompson group $V_{2,1}$ (Nekrashevych; Matui §5), which is non-amenable; purely infinite groupoids contain $\mathbb{Z}_2 * \mathbb{Z}_3$ in their full group (Matui). But $G_{[2]}$ admits no invariant probability measure: prefix replacements of unequal length scale Bernoulli measure by $2^{|p|-|q|}$, which is the modular/KMS structure of note I. So across the three algebras of the programme:

- the AF core has an invariant measure and amenable arrows;
- $\mathcal{O}_2$ has non-amenable arrows and no invariant measure;
- measure-preserving *and* non-amenable symmetries must be outer.

3 Outer expansion on the binary hierarchy: rank two

Definition 3.1 (digit interleaving). $\Phi : \{0, 1\}^{\mathbb{N}} \rightarrow \mathbb{Z}_2^2$, $\Phi(\xi) = (\sum_{k \geq 0} \xi_{2k+1} 2^k, \sum_{k \geq 0} \xi_{2k+2} 2^k)$. It is a homeomorphism carrying fair Bernoulli measure to Haar measure. The depth- m cylinder of ξ becomes $\{x \equiv a \bmod 2^{\lceil m/2 \rceil}, y \equiv b \bmod 2^{\lfloor m/2 \rfloor}\}$.

Let $A = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix}$, let τ_1, τ_2 be the unit translations, and let S_{GG} be the eight Gabber–Galil maps $A^{\pm 1}, \tau_1^{\pm 1} A^{\pm 1}, B^{\pm 1}, \tau_2^{\pm 1} B^{\pm 1}$ (signs matched; i.e. $(x, y) \mapsto (x \pm 2y, y), (x \pm (2y + 1), y), (x, y \pm 2x), (x, y \pm (2x + 1))$). All are transported to $\{0, 1\}^{\mathbb{N}}$ by Φ .

Theorem 3.2 (outer expansion on the fair hierarchy). 1. Each $s \in S_{GG}$ is a measure-preserving homeomorphism of $\{0, 1\}^{\mathbb{N}}$ mapping every depth- m cylinder partition to itself.

2. $\langle A, B \rangle \cong F_2$ (Sanov), so $\langle S_{GG} \rangle \supseteq \mathbb{Z}^2 \rtimes F_2$ is non-amenable.

3. Up to a null set, the tail relation is the $\mathbb{Z}^2$ -orbit relation, and S_{GG} normalizes it. The maps are outer symmetries. They act on the hyperfinite II_1 factor $L^\infty(X) \rtimes R_{\text{tail}}$ by trace-preserving, Cartan-preserving automorphisms. The MST shadow of A, B (for the $\mathbb{Z}^2$ -odometer with Haar measure) is induced by the matrices themselves, so $A, B \notin \llbracket R \rrbracket$, whose shadows are trivial.

4. $P_{S_{GG}}$ has spectrum in $[-1, \lambda^*]$, $\lambda^* = 5\sqrt{2}/8$, on L_0^2 , so every depth carries the same constant-degree expander.

5. The tail operators M_m (Gabber–Galil maps applied to the coordinates below depth m) satisfy $(1 - \lambda^*)(I - E_m) \leq I - M_m \leq 2(I - E_m)$ and commute with all E_k . Hence $\hat{L} = \sum_m (\lambda_{m+1} - \lambda_m)(M_m - I)$ is a constant-degree sparsifier of the PB form on the binary Cantor set itself: $(1 - \lambda^*)Q_{PB} \leq \hat{Q} \leq 2Q_{PB}$.

Proof. (1) Write $a = \lceil m/2 \rceil$, $b = \lfloor m/2 \rfloor$, so $b \geq a - 1$ and $a \geq b$. Under $(x, y) \mapsto (x \pm 2y(+1), y)$, $x \bmod 2^a$ depends only on $x \bmod 2^a$ and $y \bmod 2^{a-1}$, both of which are known. Under $(x, y) \mapsto (x, y \pm 2x(+1))$, $y \bmod 2^b$ depends on $x \bmod 2^{b-1}$, which is known. The maps are affine with unimodular linear part, hence Haar-preserving bijections. (This was also verified exhaustively at depth 10.) (2) Sanov's theorem. (3) For $x \in \mathbb{Z}_2 \setminus \mathbb{Z}$ (digits not eventually constant) and $k \in \mathbb{Z}$, $x + k$ differs from x in finitely many digits. Conversely, eventually equal digits force an integer difference. So off the null, countable union $\bigcup_{g \in \langle S_{GG} \rangle} g(\{x \in \mathbb{Z}\} \cup \{y \in \mathbb{Z}\})$, tail equivalence is $\mathbb{Z}^2$ -orbit equivalence. Affine maps with linear part in $SL_2(\mathbb{Z})$ map $v + \mathbb{Z}^2$ to $s(v) + \mathbb{Z}^2$. A normalizing, measure-preserving map induces a trace-preserving automorphism of the group-measure-space factor that preserves L^∞ . For linear s the MST cocycle is constant, $\alpha(k, v) = sk$, so $\Psi_1(s)$ is the induced map on $H^1(\mathbb{Z}^2; \mathbb{R})$. For $h \in \llbracket R \rrbracket$, $h(v) = v + c(v)$ with c continuous and bounded, so $\alpha(k, v)$ is within bounded distance of k , and $\Psi_1(h) = \text{id}$ (homomorphisms at bounded distance agree, as in MST's proof of Thm. 4.8). (4) Gabber–Galil's bound holds for $(\mathbb{Z}/m)^2$ for every m , combined with Proposition 1.2. (5) This is note III, Thm. 2.2, applied to the tail coordinates at each depth (for odd depth, x carries one more digit than y ; the tail coordinates are again a copy of $\mathbb{Z}_2^2$). $\square$

Proposition 3.3 (rank one has a trivial shadow). *For a free minimal $\mathbb{Z}$ -action (e.g. the binary odometer underlying the fair hierarchy) with ergodic invariant μ , Ψ_1 takes values in $\{\pm 1\}$ (MST Thm. 4.7 with $d = 1$). The shadow group of any family of outer symmetries is therefore finite. A non-amenable shadow group, which is the engine of Theorem 3.2, first appears at rank two, and digit interleaving manufactures that rank from a single binary hierarchy.*

Remark 3.4 (Julien–Sadun reading). The suspension of the $\mathbb{Z}^2$ -odometer is the 2-adic solenoid, the inverse limit of T^2 under $\times 2$. A and B induce homeomorphisms that are pure linear shape changes, with trivial quasi-translation and MLD parts in the JS decomposition. The expansion of Theorem 3.2 is entirely visible in their cohomological deformation class: the free subgroup of $SL_2(\mathbb{Z})$ acting on $H^1 = \mathbb{Z}[\frac{1}{2}]^2$.

Conjecture 3.5 (shadow criterion). *Let R come from a free minimal uniquely ergodic $\mathbb{Z}^d$ -action on a Cantor set with amenable $\llbracket R \rrbracket$, and let $S \subset \text{Aut}_\mu(\llbracket R \rrbracket)$ be finite, symmetric and profinite-structure preserving. If $\langle \Psi_1(S) \rangle$ is amenable, then P_S has no spectral gap. The obstacle is controlling $\ker \Psi_1 \cap \langle S \rangle$. JS's decomposition suggests that, after suspension, elements with trivial shadow are quasi-translations up to MLD/conjugacy; if this kernel is amenable, the conjecture follows from the proof of Theorem 2.1 applied to the extension. For $d \geq 2$ the hypothesis on $\llbracket R \rrbracket$ is essential: Elek–Monod give minimal $\mathbb{Z}^2$ -systems whose full group contains F_2 .*

4 The local lemma as frequency drift

Lemma 4.1 (Schur–Kesten criterion). *Let Ω be countable, and let $P = \sum_g \pi_g T_g$ on $\ell^2(\Omega)$ with $\pi_g = \pi_{g^{-1}} \geq 0$, $\sum \pi_g = 1$. Each T_g is monomial: $(T_g f)(\omega) = c_g(\omega) f(g\omega)$ with $|c_g| = 1$ and g a bijection of Ω , and P is self-adjoint. If $N : \Omega \rightarrow (0, \infty)$ satisfies*

$$\sup_{\omega} \sum_g \pi_g \sqrt{N(g\omega)/N(\omega)} \leq \lambda,$$

then $\|P\| \leq \lambda$.

Proof. $|\langle f, Pf \rangle| \leq \sum_g \pi_g \sum_{\omega} |f(\omega)| |f(g\omega)|$. Bound each product by $\frac{1}{2}(t|f(\omega)|^2 + t^{-1}|f(g\omega)|^2)$ with $t = \sqrt{N(g\omega)/N(\omega)}$. The first half sums to at most $\frac{\lambda}{2}\|f\|^2$. For the second, substitute $\eta = g\omega$;

using $\pi_g = \pi_{g^{-1}}$ it equals $\frac{1}{2} \sum_{\eta} |f(\eta)|^2 \sum_h \pi_h \sqrt{N(h\eta)/N(\eta)} \leq \frac{\lambda}{2} \|f\|^2$. Self-adjointness gives $\|P\| = \sup |\langle f, Pf \rangle| / \|f\|^2$. $\square$

Corollary 4.2 (Kesten). *For the free group F_k with uniform generators and $N(\omega) = \gamma^{2|\omega|}$, each $\omega \neq e$ has $2k - 1$ outward and one inward neighbour. Optimizing γ gives $\lambda = \sqrt{2k - 1}/k$, Kesten’s sharp bound.*

Remark 4.3 (Gabber–Galil is outward drift on the dual). Fourier transform turns P_{SGG} on $L_0^2((\mathbb{Z}/m)^2)$ into a symmetric combination of monomial operators on the nonzero frequencies. The linear parts act by A^{-T}, B^{-T} ; translations contribute phases, which Lemma 4.1 ignores. Gabber–Galil and Jimbo–Maruoka choose N from a diamond-shaped order in which, for *every* nonzero frequency, outward moves outnumber inward ones; this is the local lemma, and it gives $5\sqrt{2}/8$. Frequencies here are graded by depth exactly as in note I’s Weyl bands, and the shears preserve depth. So expansion is *outward drift within each band*: a non-amenable, tree-like dual dynamics. This is the precise form of “local neighbourhood $\Rightarrow$ global mixing”.

5 Galton–Watson duality: the network’s local lemma

For an activation at its variance fixed point, the normalized correlation map ρ is a probability generating function (squared normalized Hermite coefficients plus a bias constant), with mean $\chi_1 = \rho'(1)$ and extinction probability q . Notes I–II showed that the depth- D weight-averaged input-to-output sensitivity splits over input moment orders with weights $b_r^{(D)} = P(Z_D = r)$ for the Galton–Watson process Z with offspring law ρ . The dual action of a layer on moment orders is a branching step: each order- r “particle” is replaced by an independent sum of offspring.

Definition 5.1. The *degree- $\leq t$ leakage* at depth D is $\Lambda_t(D) = P(1 \leq Z_D \leq t)$, the share of output second moment carried by input orders $1, \dots, t$. The *input-dependence* is $\Lambda_\infty(D) = P(Z_D \geq 1)$.

Theorem 5.2 (trichotomy). 1. Subcritical ($\chi_1 < 1$, ordered): $\Lambda_\infty(D) \leq \chi_1^D$. The network forgets its input exponentially: structure wins.

2. Critical ($\chi_1 = 1$; ReLU at He initialization for every weight scale): if the offspring tail is stable of index $1 + \beta$, then $\Lambda_\infty(D) \asymp D^{-1/\beta}$ (Slack). For ReLU, $\beta = \frac{1}{2}$ and $\Lambda_\infty(D) \sim 9\pi^2/(2D^2)$. Leakage is polynomial and there is no gap in depth.

3. Supercritical ($\chi_1 > 1$, chaotic): $\Lambda_\infty(D) \rightarrow 1 - q > 0$, while for each fixed t , $\Lambda_t(D) \leq C_t \rho'(q)^D$ with $\rho'(q) < 1$ (Athreya–Ney; if $q = 0$ the rate is p_1). Every fixed-degree test of the input is fooled exponentially fast, while a fraction $1 - q$ of input randomness survives at ever higher orders. With the input as the seed, the network is a pseudorandom generator for low-degree tests.

Proof. (1) Markov’s inequality: $P(Z_D \geq 1) \leq \mathbb{E}Z_D = \chi_1^D$. (2) Slack’s theorem for critical processes in the domain of attraction of a stable law; for ReLU, $\rho(1 - x) = 1 - x + \frac{2\sqrt{2}}{3\pi}x^{3/2} + O(x^{5/2})$ (note I). (3) Standard limit theory for supercritical processes: $P(Z_D = k) \sim c_k \rho'(q)^D$ for each fixed $k \geq 1$ (Harris; Athreya–Ney, Ch. I). $\square$

network	χ_1	q	rate	$\Lambda_3(4)$	$\Lambda_3(8)$	$\Lambda_3(16)$	$\Lambda_3(32)$
tanh, $\sigma_w = 0.8, \sigma_b = 0.3$ (ordered)	0.496	1	0.496	5.8e-2	3.5e-3	1.3e-5	1.7e-10
tanh, $\sigma_w = 2.5, \sigma_b = 0$ (chaotic)	1.606	0	0.838	6.4e-1	3.4e-1	8.3e-2	5.0e-3
ReLU, He (critical)	1	1	–	2.6e-1	1.1e-1	3.1e-2	7.2e-3

Successive ratios confirm the rates: $0.496^{16} \approx 1.4 \times 10^{-5}$ against an observed 1.3×10^{-5} ; $0.838^{16} \approx 0.059$ against an observed 0.060; and $\text{ReLU} \approx (16/32)^2$.

Remark 5.3 (the analogy made exact). Gabber–Galil: a *single* frequency performs a walk whose outward drift (non-amenability) gives a spectral gap. Networks: a *branching* population of moment orders whose mean offspring decides everything.

- Supercritical networks behave like expanders: mass escapes to high frequency and low-degree tests are fooled.
- Subcritical networks contract onto structure.
- He-ReLU sits exactly on the boundary. This is the network-internal reason, complementary to Theorem 2.1, why expansion must be imported, and why note II’s seed ceiling is governed by a critical, heavy-tailed law.

6 Neural Gabber–Galil lemma (walk form)

Lemma 6.1 (collapse never closes a gap). *Let P be μ -reversible on a finite set V with spectral gap γ , and let $\pi : V \rightarrow W$ be any partition. The lumped kernel $\bar{P}(a, b) = \mu(a)^{-1} \sum_{u \in a, v \in b} \mu(u) P(u, v)$ is $\bar{\mu}$ -reversible with gap $\geq \gamma$.*

Proof. For g on W , $\mathcal{E}_{\bar{P}}(g) = \mathcal{E}_P(g \circ \pi)$ and $\text{Var}_{\bar{\mu}}(g) = \text{Var}_{\mu}(g \circ \pi)$. The gap of $\bar{P}$ is an infimum of Rayleigh quotients over a subspace of those defining γ . $\square$

Lemma 6.2 (Metropolis comparison). *If $\bar{P}$ is $\bar{\mu}$ -reversible with gap $\bar{\gamma}$ and $h > 0$ with $\bar{\mu}(h) = 1$, then $K(a, b) = \bar{P}(a, b) \min(1, h(b)/h(a))$ ($a \neq b$) is $h\bar{\mu}$ -reversible with gap $\geq \bar{\gamma} \min h / \max h$.*

Proof. $\mathcal{E}_K(g) = \frac{1}{2} \sum \bar{\mu}(a) \bar{P}(a, b) \min(h(a), h(b)) (g(a) - g(b))^2$, which is at least $\min h \cdot \mathcal{E}_{\bar{P}}(g) \geq \min h \bar{\gamma} \text{Var}_{\bar{\mu}}(g)$. Also $\text{Var}_{h\bar{\mu}}(g) \leq \int (g - \bar{\mu}g)^2 h d\bar{\mu} \leq \max h \text{Var}_{\bar{\mu}}(g)$. $\square$

Theorem 6.3 (neural Gabber–Galil lemma). *Let the source be depth $N = 2n$ of the fair binary hierarchy with the interleaved Gabber–Galil walk, of gap $\gamma \geq 1 - 5\sqrt{2}/8$ (Theorem 3.2). Let $\mathcal{C}$ be any partition of the depth- N cylinders induced by a network: continuation classes, PB ε -chain classes of any layer’s representation (note III), or activation tiles (note II). Let $h(a)$ be the class-averaged gain normalized to $\bar{\mu}(h) = 1$, restricted to classes with positive gain. Then the Metropolized lumped walk on $\mathcal{C}$ is reversible with respect to the gain-weighted class law, the network’s state seen through its own distinctions. Its spectral gap is*

$$\geq \left(1 - \frac{5\sqrt{2}}{8}\right) \cdot \frac{\min_a h(a)}{\max_a h(a)}.$$

The certificate depends on the weights only through the class-level gains. Consequently, by the expander-walk Chernoff bound (Gillman; Lezaud), for any class readout $|g| \leq 1$ the average along t steps from stationarity deviates from its gain-weighted mean by more than ε with probability $\leq 2 \exp(-c\varepsilon^2 \gamma_{\text{cert}} t)$. Each step uses 3 random bits plus one Metropolis coin.

Proof. Combine Theorem 3.2(4), Lemma 6.1 and Lemma 6.2. Concentration for reversible chains with spectral gap is Gillman/Lezaud. $\square$

Numerics ($n = 5$, so 1024 source cylinders). Cylinders are mapped to Cantor-structured inputs and pushed through a random He-ReLU network (width 256, depth 8); classes are PB ε -chain classes of the output and h is the class-averaged $\|h_L\|_1$.

ε	classes	source gap	lumped gap	gain-targeted gap	certified bound
0.02	302	0.213	0.243	0.233	0.152
0.05	88	0.213	0.297	0.281	0.155
0.10	30	0.213	0.299	0.303	0.163

Remark 6.4 (what the lemma does and does not give). The network cannot destroy expansion, only coarsen it (Lemma 6.1); weights enter only through the gain spread, which shrinks as classes coarsen with depth (note III). So expanders give *certified, randomness-efficient walks over what the network distinguishes*. They do not lower the variance floor of estimating a mean (note II’s ceiling): the mean is structure, and the walk only samples it.

7 Synthesis

source of transformations	invariant measure / amenability		expansion
inner arrows (tail, continuation, odometer)	yes	/ amenable	never (Thm. 2.1)
semigroupoid arrows (Thompson V)	$\mathcal{O}_2$,	no / non-amenable	only modular (KMS) structure
outer rank-2 symmetries (interleaved GG)	yes	/ non-amenable shadow F_2	uniform, constant degree (Thm. 3.2)
network depth, He-ReLU	–	/ critical GW	none: polynomial (Thm. 5.2)
network depth, chaotic	–	/ supercritical GW	PRG for low-degree tests
lumped onto network classes	gain-weighted		certified (Thm. 6.3)

8 Open problems

1. *Shadow criterion* (Conjecture above): characterize expansion of outer families by non-amenability of their MST/JS shadows.
2. *Expansion from the semigroupoid*. Does Thompson’s V , acting on $L^2(\{0,1\}^{\mathbb{N}})$ through the Koopman representation with its modular cocycle, have a spectral gap for some finite generating set? A yes would turn the semigroupoid arrows themselves, rather than outer symmetries, into expanders, linked to the KMS structure.
3. *Neural shadows*. Define a cohomological invariant of a network layer’s merging tree map (e.g. through the PV complex of its tile hierarchy) that extends MST/JS shadows, and decide when weight-derived maps can be outer symmetries.
4. *Supercritical seeds*. For chaotic networks, derive the analogue of note II’s lower bound: the GW law now pushes mass to high orders, so designs matter less and the ceiling changes.

A Scripts

`code/note4_checks.py`: interleaved maps preserve all cylinders; Sanov relations absent to length 8; Galton–Watson leakage for tanh and ReLU. `code/neural_gg.py`: the neural Gabber–Galil walk on network ε -chain classes.